Muhammad Jawad ur Rahman

AI Engineer & Backend Developer

+92 327 9977775 | jawdjr@gmail.com | Taxila, Pakistan | github.com/jwd-jr | linkedin.com/in/m-jawad-ur-rahman-3243b6306

ABOUT

Motivated AI Engineer & Backend Developer with hands-on experience in Python, Machine Learning, LangChain, and FastAPI, seeking a Junior AI Engineer / Backend Developer role. I build real-world AI tools that automate analysis, generate insights, and enhance learning — with strong API integration skills, experience designing database schemas and working with SQL databases, and clear communication skills for collaborating across technical and non-technical teams. Known for ability to learn quickly and contribute to impactful projects.

EDUCATION

Bachelor in Artificial Intelligence, Pak Austria Fachhochschule (PAF-IAST) *Haripur, KPK | Feb 2024 – Present*

Field: Information and Communication Technologies • EQF Level 5

SKILLS

Languages & Frameworks:

Python • FastAPI • Django • SQLAlchemy • REST APIs • API Integration • SQL Databases • Database Schemas

AI / ML:

Machine Learning • Deep Learning • NLP • Sentiment Analysis • LangChain • RAG Systems • Prompt Engineering • Hugging Face • Anthropic / Claude API • MCP • LLM Integration

Tools & Infra:

Docker • Git & GitHub • CI/CD • Vector Databases • Communication Skills

PROFESSIONAL EXPERIENCE

Backend Developer | Hoop Interactive *Wyoming, USA (Remote) | Apr 2026 – Present*

- Built and maintained backend REST APIs using Python and FastAPI for production-level applications, supporting API integration across multiple product features
- Managed SQL databases using SQLAlchemy with clean, normalized schema design
- Integrated AI models into web applications, extending product capabilities without added latency
- Containerized services with Docker and supported CI/CD workflows to streamline deployment
- Collaborated cross-functionally on testing, debugging, and new feature development, communicating progress clearly with the team

AI Engineer | Ctrl Alt Crew *Islamabad, Pakistan | Oct 2025 – Mar 2026*

- Developed and optimized ML models using TensorFlow, Scikit-learn, and PyTorch, evaluating multiple model variants to select the best-performing approach based on validation accuracy
- Conducted EDA and prepared real-world datasets for model training across several projects, standardizing preprocessing steps to reduce turnaround time
- Built and integrated RAG systems using LangChain for enhanced AI responses
- Developed NLP pipelines for text processing, analysis, and language understanding
- Assisted in deploying models to test environments, writing technical documentation, and communicating results to the team

PROJECTS

AI-Powered Data Analysis & Visualization Tool | Python • LangChain • FastAPI • Plotly

- Built an end-to-end automated data analysis pipeline from a single dataset upload, cutting manual analysis time significantly for exploratory tasks
- Implemented data cleaning, outlier detection, and interactive visualizations (histograms, scatter plots, correlations)
- Integrated LLMs to automatically summarize patterns, trends, and key insights

AI-Based Self-Study & Auto-Quiz Generator | LangChain • FastAPI • PDF Parsing

- Converted lecture slides into structured learning material with auto-generated quizzes, reducing manual study-prep time

- Built a FastAPI backend for file uploads, content processing, and API handling
- Used embeddings and prompt chaining to produce accurate educational output

YouTube Comment Analyzer — Live Chrome Extension | FastAPI • Scikit-learn • K-Means • Groq AI • Railway

- Built and deployed a full-stack ML app: backend live on Railway, frontend as a working Chrome Extension
- Sentiment analysis on up to 1,000 YouTube comments using 5 trained ML models (Logistic Regression, Naive Bayes, Random Forest, SVM, AdaBoost); best model served via FastAPI
- Opinion clustering with K-Means + TF-IDF to surface top 3 majority opinions, auto-labeled via Groq AI with percentage breakdown
- Results displayed as an interactive donut chart in the Chrome Extension popup using Chart.js

Tax Filing System — Full Stack Project | Python • FastAPI • SQL • SQLAlchemy • ERD • UML

- Built a complete tax filing system from scratch, covering the full development lifecycle from system design to implementation
- Designed class diagrams and ERDs before coding, then implemented a fully normalized relational database
- Developed REST APIs using FastAPI to handle tax records, user data, and file management operations
- Implemented core features including tax record submission, data storage, and transaction tracking

LANGUAGES

Chitrali (Native) • Urdu (Native / C2) • English (Proficient / B2–C2)